

Assignment 3

Day4 Component-Specific Troubleshooting Practical(1 hour):
Identify fix issues in faulty components (RAM, GPU, Storage).
Take: Create a checklist for diagnosing hardware issues.

Task 1: RAM Reseating

To remove and re-insert the RAM module to simulate RAM reseating and check whether the system works properly.

Steps Performed:

1. Shut down the computer completely.
2. Disconnect the power cable/charger.
3. Open the computer case or RAM access panel.
4. Locate the RAM slot on the motherboard.
5. Release the clips on both sides of the RAM module.
6. Carefully remove the RAM module.
7. Check the RAM and slot for visible dust or damage.
8. Re-insert the RAM firmly into the slot until the clips lock.
9. Close the case/panel and reconnect the power.
10. Start the computer and check whether it boots normally.

Before reseating, the system may show no display, boot problems, or beep sounds. After properly reseating the RAM, the computer should normally boot if loose RAM was the problem.

Result:

RAM was successfully reseated and the system was checked for proper operation.



Fig 1: RAM slots on motherboard
(inside desktop PC)

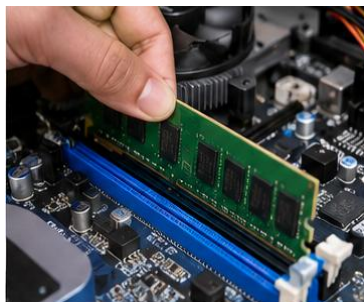


Fig 2: Removing the RAM module

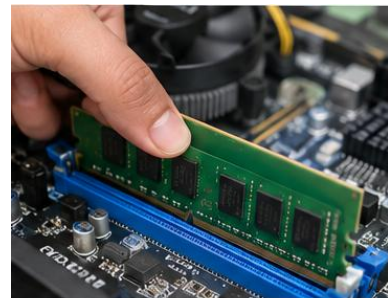


Fig 3: Re-inserting the RAM module

Task 2: GPU Troubleshooting Checklist

To diagnose and troubleshoot a faulty GPU in a gaming PC.

No.	Troubleshooting Step	What to Check
1	Identify the symptom	Check for no display, screen flickering, artifacts, or crashes
2	Check display cable	Make sure HDMI/DisplayPort cable is connected properly
3	Check GPU power	Verify that required GPU power connectors are firmly connected
4	Check GPU installation	Make sure the GPU is properly seated in the PCIe slot
5	Check drivers	Update or reinstall the graphics driver
6	Check temperature	Monitor GPU temperature during operation
7	Test the GPU	Run a suitable graphics/game test and check stability
8	Confirm the fix	Verify that display and gaming work normally without crashes



Fig 4: Dedicated GPU installed in a gaming PC



Fig 5: GPU power connection (PCIe)



Fig 6: DisplayPort/HDMI connected to GPU

The GPU troubleshooting checklist helps identify whether the problem is caused by a cable, connection, driver, temperature, or GPU hardware

Task 3: Storage Issue Simulation

To simulate a storage connection problem by disconnecting and reconnecting a SATA/NVMe drive.

Steps:

1. Shut down the test computer.
2. Disconnect the power supply.
3. Open the computer case.
4. Locate the SATA SSD/HDD or NVMe SSD.
5. Disconnect the SATA data/power cable, or remove the NVMe drive if the test system permits it.
6. Start the computer and check the BIOS/UEFI.
7. Observe whether the storage drive is detected.
8. Shut down the computer again.
9. Reconnect the drive properly.
10. Start the computer and check BIOS/UEFI again.
11. Boot into Windows and check whether the drive is available.

Before Reconnecting: Possible observations:

- BIOS may not show the storage drive.
- The operating system may show a boot-device error if the disconnected drive contains the OS.
- A secondary drive may simply disappear from Windows.

After Reconnecting:

- BIOS should detect the drive again.
- Windows should normally recognize the storage device.
- If it contains the operating system, the PC should be able to boot normally.

The storage connection was checked before and after reconnection, and the drive detection status was compared.



Fig 7: SATA SSD installed in PC



Fig 8: SATA data and power connections



Fig 9: Storage drive detected in BIOS/UEFI



Fig 10: No boot device error (before reconnecting)

Task 4: PC Randomly Restarts

A PC randomly restarts while being used.

Component I would check first: RAM

Reason:

Among the listed components, RAM is a reasonable first check because a loose or faulty RAM module can cause system instability, crashes, or unexpected restarts. It is also relatively straightforward to reseal and test.

Diagnostic Steps:

1. Shut down the PC and disconnect the power.
2. Open the computer case.
3. Locate the RAM modules.
4. Remove and carefully reseal the RAM.
5. Check whether the RAM is properly locked into the slot.
6. If there are multiple RAM modules, test them one at a time where practical.
7. Run a memory diagnostic test such as **Windows Memory Diagnostic**.
8. Use the PC normally and observe whether the random restarting continues.
9. If the problem remains, continue checking the GPU, storage, power supply, temperatures, and other hardware.

RAM was selected as the first component to check because an improper RAM connection or memory fault can cause system instability. Further components should be tested if the problem continues.

Checklist for Diagnosing Hardware Issues

- ☐ Identify the symptoms
- ☐ Check physical connections
- ☐ Reseat the suspected component
- ☐ Check drivers and system settings
- ☐ Run appropriate diagnostic tests
- ☐ Check temperatures
- ☐ Test the system after the repair
- ☐ Record the final result